From Churn Prediction to Retention Decisions: An Imbalance- and Threshold-Aware Machine Learning Framework for Telecom Customer Churn

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Abstract—Customer churn prediction is useful only when predicted risk can be translated into a sensible retention decision. Most telecom churn studies report accuracy, F1-score or ROC-AUC, yet the operational consequences of a probability threshold are often less visible. This study develops a leakage-safe framework that examines class imbalance, probability thresholds, decision consequences and model explanations within one experimental design. The IBM Telco Customer Churn dataset, containing 7,043 customers, was evaluated with five machine-learning algorithms and four imbalance-handling strategies. Average Precision (AP) was used as the primary ranking metric under stratified five-fold cross-validation. Tuned candidates were then evaluated using leakage-free out-of-fold probabilities, threshold scans and three transparent false-negative to false-positive scenarios (2:1, 5:1 and 10:1). A minimax relative-regret rule selected XGBoost trained on the original data with a threshold of 0.18. On the untouched test set, the model obtained AP 0.665 and ROC-AUC 0.847. Lowering the threshold from 0.50 to 0.18 increased churn recall from 0.532 to 0.861 and reduced missed churners from 175 to 52, while false positives increased from 97 to 380. The selected threshold reduced normalized decision cost by 34.16% and 51.27% under the 5:1 and 10:1 scenarios, respectively, but was 8.28% worse under 2:1. SHAP analysis identified Contract, tenure, InternetService, MonthlyCharges and OnlineSecurity as the strongest model-level contributors. The results show that churn models should be judged not only by ranking quality, but also by the decisions created by their operating threshold.

Keywords—*customer churn; class imbalance; decision threshold; Average Precision; XGBoost; explainable AI*

I. INTRODUCTION

Customer churn is a persistent concern in telecommunications because losing an established subscriber affects recurring revenue, service planning and the cost of retention campaigns. Machine-learning methods are now routinely used to estimate churn risk from customer demographics, subscribed services, billing patterns and account history. Recent studies have compared logistic regression, decision trees, random forests, gradient boosting and neural models, while others have introduced resampling, ensemble learning and explainable artificial intelligence (Chang et al., 2024; Alotaibi and Haq, 2024; Asif et al., 2025).

The central difficulty, however, is not simply to obtain a model with a high score. Telecom churn data are commonly imbalanced, and the default probability threshold of 0.50 can hide a large number of missed churners. A model may rank customers well and still produce an operating point that is poorly aligned with retention priorities. Manzoor et al. (2024), after reviewing 212 churn studies, noted that profitability-oriented evaluation remains under-researched and argued for stronger links between predictive modelling and business decision making. Poudel et al. (2024) similarly showed that reasonable overall performance can coexist with a meaningful number of false negatives, which is important when the objective is early retention action.

Class imbalance has been approached through class weighting, synthetic oversampling and hybrid resampling. Ouf et al. (2024) and Wagh et al. (2024), for example, investigated SMOTE-ENN-based strategies, whereas Sikri et al. (2024) examined churn behaviour under different class ratios. Explainability has also become more common through SHAP and LIME (Chang et al., 2024; Asif et al., 2025). These developments are valuable, but they are usually reported as separate modelling choices. Within the reviewed literature, there is limited evidence of a single leakage-safe evaluation process that connects imbalance handling, threshold selection, false-negative/false-positive consequences and post-hoc explanation.

This study addresses that gap by treating churn prediction as a decision problem rather than a leaderboard problem. The work asks three questions. RQ1 examines how imbalance strategies change model performance under the same validation protocol. RQ2 studies how probability thresholds alter the balance between detected churners and unnecessary interventions, and identifies one model-threshold pair that remains robust across several decision-cost assumptions. RQ3 investigates which customer attributes most strongly influence the final model at both global and individual levels.

The contribution is therefore not a new classifier, resampling algorithm or explainability method. Instead, the study

integrates these components into a retention-oriented framework: leakage-safe preprocessing, controlled imbalance experiments, AP-based model ranking, out-of-fold threshold analysis, transparent scenario costs, minimax relative-regret selection and SHAP interpretation. The remainder of the paper reviews related work, describes the methodology, reports the experimental results, discusses their implications and closes with the main findings and limitations.

II. RELATED WORK

Customer churn has been studied with a broad range of machine-learning approaches. Manzoor et al. (2024) provide a comprehensive view of the field, covering preprocessing, feature selection, modelling and evaluation, and identify profit-aware evaluation and explainability as important directions. In application studies, Chang et al. (2024) combined telecom churn classification with SHAP and LIME, while Poudel et al. (2024) used tabular machine-learning models and both global and local SHAP explanations. These studies demonstrate that predictive performance and interpretability can be considered together.

Imbalance handling is another recurring theme. Sikri et al. (2024) analysed different class ratios and reported XGBoost results with accuracy 89.60%, precision 76.04%, recall 62.82% and F1-score 68.80% for one configuration. Ouf et al. (2024) integrated XGBoost with SMOTE-ENN and reported large changes in classifier behaviour after resampling. Wagh et al. (2024) also examined the IBM Telco data with SMOTE-ENN and survival analysis. These findings make clear that resampling can materially alter the minority-class operating point, but they also underline the need for leakage-safe and comparable evaluation.

Several studies have focused on model comparison and optimisation. Saleh and Saha (2023) compared multiple classifiers across telecom datasets, including the IBM data. Alotaibi and Haq (2024) evaluated decision tree, logistic regression, XGBoost, LightGBM, ANN and an averaging ensemble; the ensemble did not automatically outperform the stronger individual models. AlShourbaji et al. (2023) combined gradient boosting with metaheuristic optimisation, oversampling and feature selection. More recently, Asif et al. (2025) proposed an explainable ensemble of XGBoost, CatBoost and LightGBM on a dataset exceeding two million records and used SHAP and LIME to interpret predictions.

Taken together, the literature already contains strong work on classifiers, ensembles, imbalance handling, tuning and XAI. The narrower gap addressed here is the transition from probability ranking to retention action: how training-time imbalance choices interact with threshold selection, how many churners are missed or how many non-churners are unnecessarily targeted, and how one operating point can be selected when the exact relative cost of these errors is uncertain.

TABLE I. SUMMARY OF CLOSELY RELATED STUDIES

Study	Main approach	Relevance to this study
Manzoor et al. (2024)	Review of 212 churn studies	Highlights limited profit-oriented evaluation and need for decision support.
Chang et al. (2024)	ML models + SHAP/LIME	Shows value of explainable churn prediction.
Sikri et al. (2024)	Class-ratio analysis + XGBoost	Demonstrates sensitivity of churn metrics to imbalance treatment.
Poudel et al. (2024)	Tabular ML + SHAP	Shows strong overall metrics can still leave missed churners.
Saleh and Saha (2023)	Multi-dataset classifier comparison	Supports cross-model evaluation on telecom churn data.
Wagh et al. (2024)	SMOTE-ENN + ML + survival	Illustrates hybrid balancing for IBM Telco churn.
Asif et al. (2025)	TriBoost ensemble + SHAP/LIME	Combines strong prediction with XAI on large telecom data.
Ouf et al. (2024)	XGBoost + SMOTE-ENN	Shows large changes after hybrid resampling.
Alotaibi and Haq (2024)	ML, ANN, ensemble, tuning	Comparable short experimental study on IBM Telco data.
AlShourbaji et al. (2023)	Boosting + metaheuristic optimisation	Represents optimisation-focused churn modelling.

III. METHODOLOGY

The experimental design was organised so that every modelling choice was made on development data and the final test set remained locked. Figure 1 shows the complete workflow, from data audit to final paper outputs. The main methodological stages are described below.

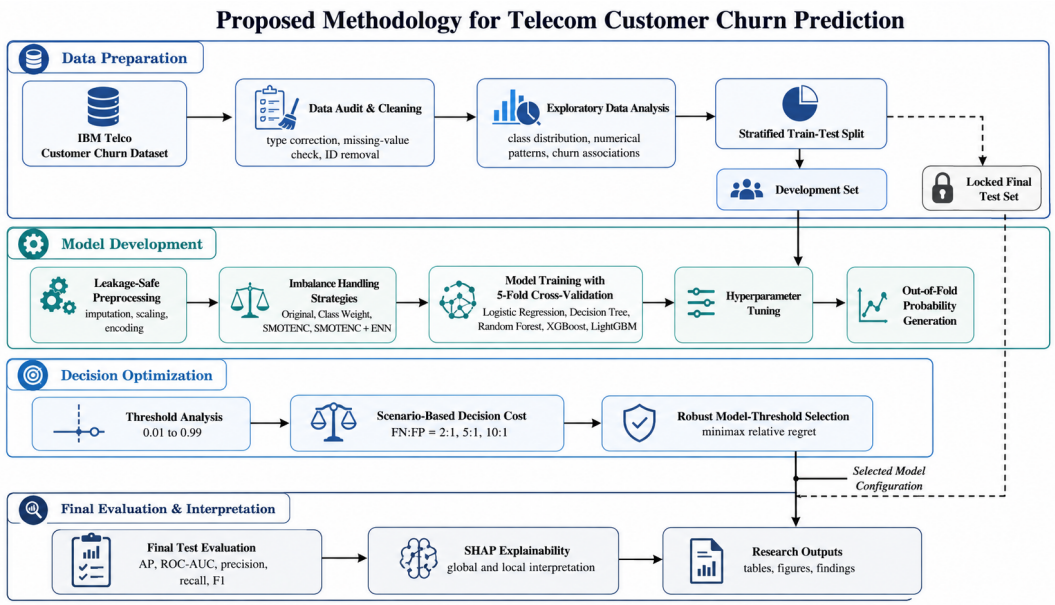


Figure 1: Proposed leakage-safe, imbalance-aware, threshold-aware, and decision-oriented telecom churn prediction framework.

A. Dataset and Data Preparation

The IBM Telco Customer Churn dataset contains 7,043 customers and 21 original columns. The target distribution is imbalanced: 5,174 customers (73.46%) are non-churners and 1,869 (26.54%) are churners. The unique customerID field was removed. TotalCharges, initially stored as an object, contained 11 blank entries; these were converted to missing numeric values and retained for fold-wise imputation. Churn was encoded as 1 for Yes and 0 for No. No manual feature engineering was introduced.

Exploratory analysis showed clear descriptive differences between churners and non-churners. Median tenure was approximately 10 months for churners and 38 months for non-churners. Higher churn rates were observed among month-to-month contracts, fibre-optic users, electronic-check customers and customers without online security or technical support. These patterns were treated as associations and were not interpreted as causal effects.

B. Locked Hold-out and Leakage-Safe Preprocessing

A stratified 80/20 split with random_state = 42 created a development set of 5,634 customers and a locked test set of 1,409. The development set contained 4,139 non-churners and 1,495 churners; the test set contained 1,035 and 374, respectively. All model comparison, preprocessing, tuning and threshold selection used development data only.

Three variables (tenure, MonthlyCharges and TotalCharges) were treated as continuous and processed with median imputation followed by standardisation. The remaining 16 predictors, including SeniorCitizen, were treated as categorical or binary. They were imputed with the most frequent category and ordinal-encoded before any sampling step. One-hot encoding was applied after sampling. This ordering allowed SMOTENC to operate on categorical-aware representations rather than interpolating one-hot dummy variables. Every transformation was fitted inside the training portion of each fold.

C. Models, Imbalance Strategies and Validation

Five classifiers were evaluated: Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), XGBoost and LightGBM. Four imbalance strategies were compared: Original data, Class Weight, SMOTENC and SMOTENC followed by Edited Nearest Neighbours (ENN). In the hybrid pipeline, ENN used n_neighbors = 3 and sampling_strategy = "all". The design produced 20 baseline model-strategy combinations evaluated on the same stratified five-fold splits.

Average Precision (AP) was the primary model-ranking metric because it summarises precision-recall behaviour in an imbalanced binary problem. ROC-AUC, precision, recall, F1-score, accuracy, false negatives (FN) and false positives (FP) were retained as secondary measures. The AP leader for each model was carried forward, together with selected near-best strategies where AP loss was no greater than 0.005 and recall improved materially. Seven candidates were then tuned by randomised search. In total, 205 parameter configurations were evaluated over five folds (1,025 fits).

D. Out-of-Fold Probabilities and Threshold Analysis

Leakage-free out-of-fold (OOF) probabilities were generated for the seven tuned candidates. Each development customer was predicted only by a model that had not been trained on that customer. These OOF scores, rather than the final test set, were used to study operating thresholds.

Thresholds from 0.01 to 0.99 were evaluated in steps of 0.01. For each threshold, precision, recall, F1, FN and FP were recorded. The search range was extended after an initial boundary check showed that one high-FN-cost optimum occurred at the lower edge of the first scan. No test information was used during this extension.

E. Decision-Cost Framework

The dataset does not contain observed customer lifetime value, retention-offer cost or profit margin. For that reason, the study does not claim monetary profit. Instead, three transparent scenarios were defined in which one missed churner was assumed to have two, five or ten times the consequence of one unnecessary intervention. For threshold t and relative FN cost r, normalized decision cost was defined as:

$$C_r(t) = \frac{r \cdot FN(t) + FP(t)}{N}$$

where N is the number of evaluated customers and r
in {2, 5, 10}. Lower values indicate a more favourable decision under that assumed scenario.

F. Robust Model-Threshold Selection

Different cost scenarios naturally favoured different thresholds. To select one operating point without pretending that a single cost ratio was known, relative regret was computed against the best achievable cost within each scenario:

$$R_{m,t,s} = \frac{C_{m,t,s} - C_s^*}{C_s^*}$$

The final combination minimised the largest relative regret across the three scenarios:

$$(m^*, t^*) = \arg\min_{m,t} \max_{s} R_{m,t,s}$$

Ties were resolved using mean relative regret, mean normalized cost and then OOF predictive quality. This procedure selected XGBoost trained on the original distribution with threshold 0.18. The tuned model used 200 estimators, learning rate 0.05, max_depth 2, min_child_weight 5, subsample 0.8, colsample_bytree 0.8, reg_alpha 0 and reg_lambda 1.0.

G. Final Evaluation and SHAP

After the model and threshold were locked, the pipeline was refitted on all development data and evaluated once on the untouched test set. SHAP TreeExplainer was then applied to all 1,409 test customers. Encoded SHAP values belonging to the same original feature were summed per customer before computing mean absolute importance, preventing multi-level categorical variables from being inflated simply because they produced more dummy columns. SHAP was used only for explanation after model selection and was not treated as evidence of causality.

IV. EXPERIMENTAL RESULTS

A. RQ1: Effect of Class-Imbalance Handling

The tuned results show that imbalance handling changed the operating behaviour of the models more than it changed their ranking quality. XGBoost on the original distribution achieved the highest mean CV AP (0.6714), closely followed by class-weighted XGBoost (0.6700). The class-weighted model, however, produced much higher recall at the

default threshold. LightGBM and Decision Tree with SMOTENC+ENN also achieved high recall, but at lower AP and precision.

TABLE II. TUNED CANDIDATE COMPARISON ON FIVE-FOLD DEVELOPMENT CV

Candidate	Strategy	AP	ROC-AUC	Precision	Recall	F1
XGBoost	Original	0.6714	0.8503	0.6683	0.5298	0.5908
XGBoost	Class Weight	0.6700	0.8498	0.5212	0.8161	0.6360
Random Forest	Original	0.6666	0.8480	0.6768	0.5050	0.5781
Logistic Reg.	Original	0.6616	0.8462	0.6531	0.5438	0.5928
Logistic Reg.	Class Weight	0.6600	0.8460	0.5174	0.8007	0.6284
LightGBM	SMOTENC +ENN	0.6587	0.8438	0.4956	0.8321	0.6211
Decision Tree	SMOTENC +ENN	0.5802	0.8207	0.4815	0.8140	0.6048

Pooled OOF AP remained similarly close for XGBoost Original (0.6682) and XGBoost Class Weight (0.6669). Yet, at threshold 0.50, their recalls were 0.530 and 0.816, respectively. When the operating point was matched at roughly 80% recall, the difference almost disappeared: XGBoost Original at threshold 0.27 produced recall 0.802 and precision 0.523, while class-weighted XGBoost at 0.51 produced recall 0.804 and precision 0.525. This result answers RQ1: imbalance treatment is model-dependent, and its apparent benefit at a fixed threshold can overlap with what is achievable through threshold adjustment.

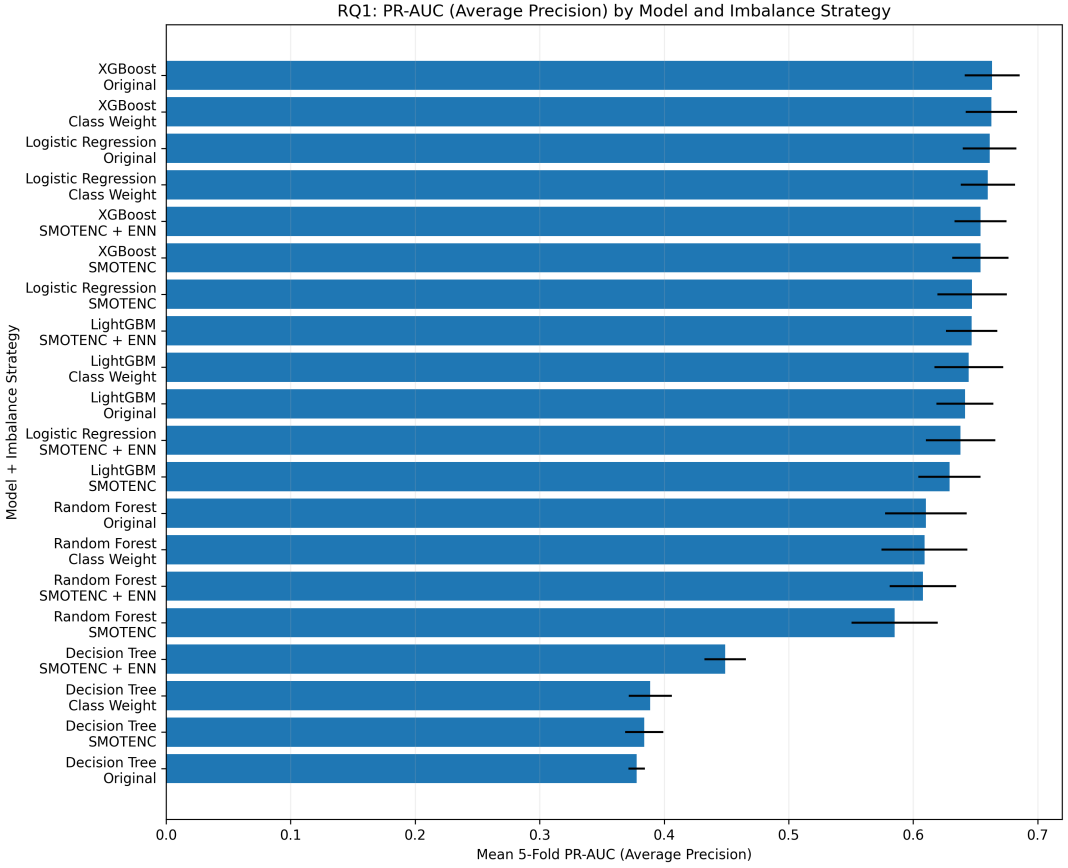


Figure 2: AP comparison across the model and imbalance experiments.

B. RQ2: Threshold and Decision Consequences

The robust OOF selection chose XGBoost Original at threshold 0.18. Its OOF AP and ROC-AUC were 0.6682 and 0.8496. At the selected threshold, OOF recall reached 0.8823 with precision 0.4677 and F1 0.6114. The maximum relative regret across the three cost scenarios was 0.1431, lower than the competing model-threshold combinations.

On the locked test set, AP was 0.6650 and ROC-AUC was 0.8468. The small changes from OOF (–0.0032 AP and –0.0028 ROC-AUC) indicate that ranking performance transferred closely to unseen data. More importantly, the operating threshold changed the error profile. At 0.50, recall was 0.5321 with 175 false negatives. At 0.18, recall rose to 0.8610 and false negatives fell to 52. The cost of that gain was a rise in false positives from 97 to 380 and a reduction in overall accuracy.

TABLE III. FINAL LOCKED-TEST PERFORMANCE AT DEFAULT AND SELECTED THRESHOLDS

Threshold	Accuracy	Precision	Recall	F1	FN	FP
0.50	0.8070	0.6723	0.5321	0.5940	175	97
0.18	0.6934	0.4587	0.8610	0.5985	52	380

Thus, 123 additional churners were captured, but 283 additional non-churners were flagged. This is the central trade-off of the study: a lower threshold is not simply 'better'; it is preferable only when the cost of missing churners is sufficiently high relative to the cost of unnecessary retention contact.

TABLE IV. FINAL NORMALIZED DECISION COST

FN:FP scenario	Cost at 0.50	Cost at 0.18	Change
2:1	0.317246	0.343506	8.28% worse
5:1	0.689851	0.454223	34.16% lower
10:1	1.310859	0.638751	51.27% lower

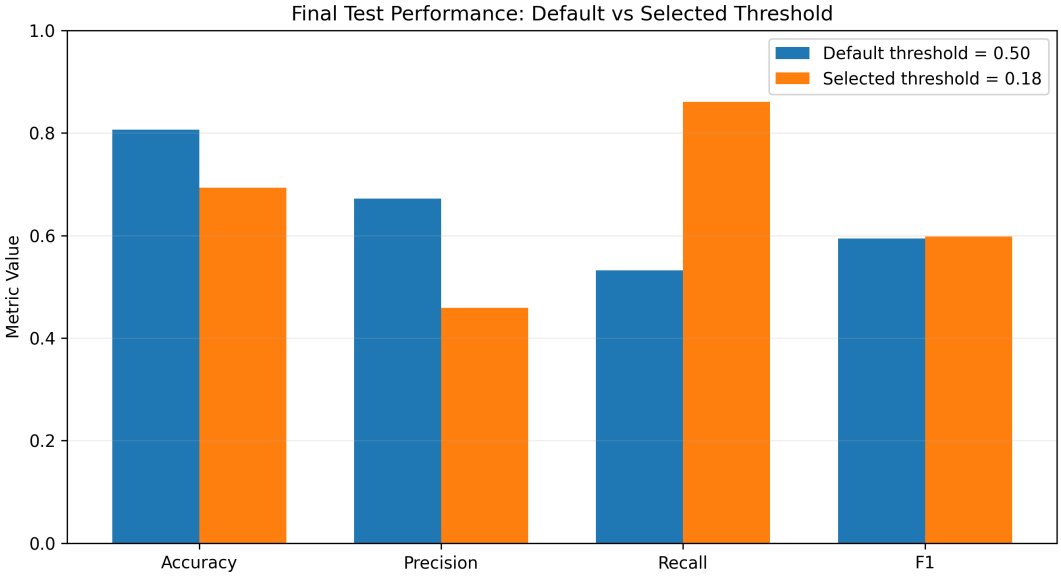


Figure 3: Default versus selected threshold and decision-cost comparison.

C. RQ3: Explainability of the Final Model

The grouped SHAP analysis identified Contract as the most influential original feature (mean |SHAP| = 0.6622), followed by tenure (0.5118), InternetService (0.2908), MonthlyCharges (0.2298) and OnlineSecurity (0.2215). PaymentMethod, TechSupport, PaperlessBilling, MultipleLines and TotalCharges completed the top ten. At the encoded level, month-to-month contract, tenure, fibre-optic internet, MonthlyCharges, lack of OnlineSecurity, two-year contract, electronic check and lack of TechSupport were among the strongest contributors.

Local explanations also showed why threshold choice matters for individual customers. A high-confidence true positive received churn probability 0.9187, while a borderline non-churner and one false-negative churner received 0.1801 and 0.1795, respectively, placing them on opposite sides of the selected decision boundary. SHAP therefore helps explain how the model arrived at a risk score, but it does not establish that changing one attribute would causally prevent churn.

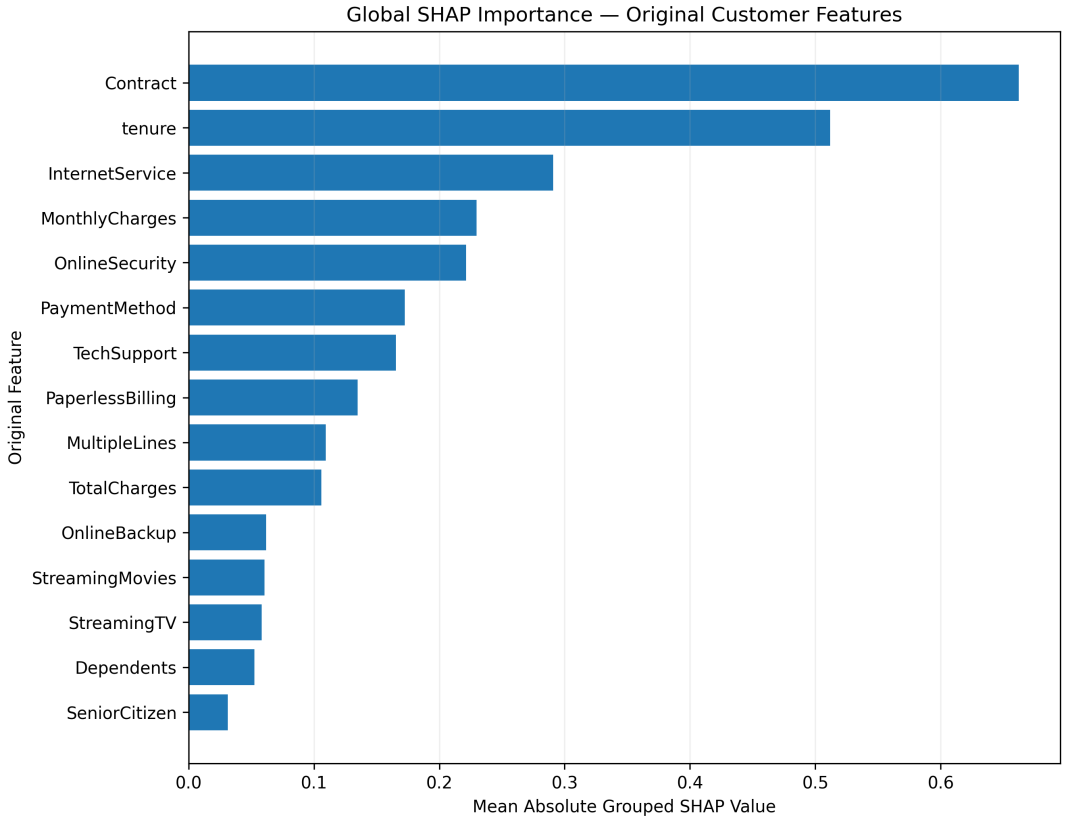


Figure 4: Global grouped SHAP feature importance for the final XGBoost model.

V. DISCUSSION

Three observations stand out from the experiments. First, the choice of imbalance strategy should not be judged at threshold 0.50 alone. Class weighting substantially increased default-threshold recall, but the corresponding AP remained almost unchanged relative to XGBoost trained on the original data. Once thresholds were aligned to comparable recall, the two XGBoost variants behaved almost identically. This suggests that training-time imbalance handling and post-training threshold selection are partly substitutable and should be analysed together rather than reported as separate improvements.

Second, the final threshold exposes the practical cost of higher churn sensitivity. At 0.18, the model detected 86.1% of test churners, compared with 53.2% at 0.50. That gain reduced missed churners from 175 to 52, but it also generated 380 false positives. Under the 2:1 scenario this was not worthwhile; under 5:1 and 10:1 it produced sizable cost reductions. The result is therefore deliberately conditional. No threshold is universally optimal. A retention team should choose an operating point only after deciding how it values a missed churner relative to an unnecessary intervention.

Third, the selected model remained interpretable at both portfolio and customer level. Contract and tenure dominated the SHAP ranking, while internet service, billing and support variables also contributed strongly. These patterns are consistent with the broad direction reported in explainable telecom churn studies (Chang et al., 2024; Poudel et al., 2024), but the present analysis keeps a clear distinction between model explanation and causality.

The study has several limitations. It uses one public dataset and therefore does not establish external validity across operators or countries. The decision-cost ratios are hypothetical because the data contain no observed customer lifetime value, retention-offer cost, campaign response or profit margin. Probability calibration was not separately

optimised, and the scores should be treated as risk scores for ranking and thresholding rather than guaranteed frequencies. External validation and prospective retention experiments were also outside the scope of the work. Future studies could incorporate real customer value and intervention costs, test calibrated probabilities, validate on additional operators and compare a global threshold with customer-specific treatment policies.

VI. CONCLUSION

This study presented a leakage-safe, imbalance-aware and threshold-aware framework for telecom churn prediction. Five machine-learning models and four imbalance strategies were compared, tuned and evaluated with AP as the primary ranking measure. OOF probabilities were then used to study thresholds and transparent FN:FP cost scenarios without touching the final test set.

The final choice was XGBoost trained on the original distribution with threshold 0.18. On the locked test set it achieved AP 0.6650 and ROC-AUC 0.8468. Relative to threshold 0.50, recall rose from 0.5321 to 0.8610 and false negatives fell from 175 to 52, although false positives increased from 97 to 380. The selected operating point reduced normalized cost under the 5:1 and 10:1 scenarios but not under 2:1. The main conclusion is therefore straightforward: a strong churn model is not enough; its threshold must be judged against the consequences of the decisions it creates. SHAP further identified the customer attributes most influential to the final model and provided a transparent layer for interpreting individual risk scores.

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